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SUPERSEDING
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October 3, 1961

FEDERAL SPECIFICATION

PLASTIC SHEET AND FILM, POLYSTYRENE, BIAXIALLY ORIENTED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers biaxially oriented polystyrene sheet and film of 0.00075- to 0.020-inch nominal thickness.

1.2 Classification.

1.2.1 *Types.* Biaxially oriented polystyrene sheet and film covered by this specification shall be of the following types, as specified:

Type I.—Unmodified polystyrene
Type II.—Styrene-acrylonitrile copolymer

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-268 — Paper, Kraft, Untreated, Wrapping.
UU-P-271—Paper, Wrapping, Waterproofed Kraft.
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-D-723—Drums, Fiber.
PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof — for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 1—Standard for Laboratory Atmospheric Conditions for Testing.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 406—Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

American Society for Testing and Materials (ASTM) Standards:

D 1504-61—Method of Test for Orientation Release Stress of Plastic Sheeting.
D 1505-60T—Density of Plastics by the Density-Gradient Technique.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Type I.* Type I sheet and film shall be made from unmodified polystyrene resin with or without the addition of colorants and small amounts of lubricants. The finished product shall have molecular orientation.

3.1.2 *Type II.* Type II sheet and film shall be made from styrene-acrylonitrile copolymer resin with or without the addition of colorants and small amounts of lubricants. The finished product shall have molecular orientation.

3.2 Form. The material shall be furnished as cut sheets or as rolls, as specified by the procuring activity. Roll shipments shall contain an average of not more than three pieces per roll and no piece shall be less than 200 feet in length.

3.3 Dimensions. The length, width, and thickness of the sheet or film shall be as specified in the contract or purchase order.

3.4 Tolerances. The tolerance for thickness shall be as specified in table I. Tolerance for roll widths up to 30 inches shall be ± 3/16 inch. For roll widths from 30 to 50 inches, the tolerance shall be ± 3/8 inch; and for widths from 50 to 60 inches, the tolerance shall be ± 1/2 inch. Standard size trimmed sheets shall be 20 inches ± 1/16 inch by 50 inches ± 1/16 inch.

TABLE I. *Standard nominal thickness and tolerances on thickness*

Nominal thickness, inch	Actual thickness	
	Minimum, inch	Maximum, inch
0.00075	0.0006	0.0009
.001	.0008	.0012
.00125	.0010	.0016
.0015	.0012	.0018
.002	.0017	.0023
.003	.0025	.0035
.005	.0042	.0058
.0075	.0064	.0086
.01	.0085	.0115
.015	.0128	.0172
.020	.017	.023

3.5 Color and transparency. The color and transparency or opacity of the sheet or film shall be as specified in the contract or purchase order.

3.6 Finish. The sheet or film shall be furnished with a high polish[1] finish, or a spe-

[1] Polish describes surface smoothness in terms of light reflection, generally mirrorlike, but with localized distortion of reflected images.

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TABLE II. *Physical properties of polystyrene*

Property	Type I	Type II
Orientation release stress, min., pounds per square inch (p.s.i.)	75	75
Orientation ratio, each direction, min.	2:1	2:1
Density[6], D23°C., g. per ml.:		
Minimum	1.045	1.065
Maximum	1.07	1.085
Specific gravity [1, 5], 23°/23°C.:		
Minimum	1.045	1.065
Maximum	1.07	1.085
Water absorption[2, 5], 24 hours immersion, maximum, percent	0.05	0.5
Tensile strength[3], minimum, p.s.i	8000	10,500.
Elongation at rupture[3], minimum, percent	3.0	15
Dissipation factor[4], 1 megacycle, maximum	0.0005	0.01
Dielectric constant[4], 1 megacycle, maximum	2.70	3.1
Luminous transmittance[4], percent, minimum	88	88
Water vapor permeability[6], grams, per square meter, per 24 hours, per mm. thickness, per cm. Hg at 25°C., max.	0.5	0.5

[1] Applicable only to unpigmented material.

[2] Specimens conditioned in an oven at 50° ± 3°C. (122° ± 5.4°F.) for 24 hours.

[3] Specimens conditioned at 23°C. (73.5°F.) and 50 percent relative humidity; 1/8 inch or thinner specimens for 48 hours.

[4] Applicable only to clear colorless material.

[5] Not applicable to thin films, 0.003 inch thickness or less.

[6] Applicable only to thin films, 0.003 inch thickness or less.

cial finish, as specified in the contract or purchase order.

3.7 **Chemical resistance.** The resistance of the material to specific chemicals shall be as specified in the contract or purchase order.

3.8 **Physical properties.** Test specimens cut from sheet or film and tested as specified in section 4 shall meet the physical properties listed in table II. For the dielectric constant and dissipation factor test, 1/8-inch specimens shall be prepared from the same material, under production conditions similar as possible to the production conditions of the sheet or film under test. No special treatment shall be used that will tend to improve the properties of the specimens as compared with those of the sheet or film supplied under this specification.

3.9 **Fog resistance.** When specified, the sheet or film shall be treated on one side in such manner as to provide fog resistance.

3.10 **Workmanship.** The sheet and film shall be commercially free from warpage, cracks, scratches, and other defects that affect the appearance or that might affect the serviceability. The sheet and film shall be uniform in color, transparency or opacity, finish, density, and other physical properties.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling.

4.2.1 *Lot.* A lot shall consist of material produced by one manufacturer, subjected to the same manufacturing and processing conditions. Each lot shall contain material of one type only.

4.2.2 *Sampling for inspection.* Sampling shall be accomplished in accordance with MIL-STD-105.

4.3 Inspection.

4.3.1 *Inspection of preparation for delivery.* The packaging, packing, and marking of the plastic sheet or film shall be examined to determine conformance with the requirements of section 5 of this specification and as specified in table III. Examinations of packaging, packing, and marking requirements not covered by the referenced specifications shall be in accordance with MIL-STD-105.

TABLE III. *Inspection of preparation for delivery*

Examination	Defect
Marking—shipping container, interior and intermediate containers, as applicable	Incorrect, omitted, illegible; improper size, location, sequence, or application.
Materials	Nonconforming as to class, grade, style, or other category of the specification. Required component missing, damaged, or otherwise defective.
Workmanship and preservation	Inadequate application of components, such as poor closure of case liners or container flaps, loose or over-tightened strapping, sharp ends on strapping, inadequate stapling, protruding nails, bulging of containers, insufficient dunnage or blocking and supports.
Contents	Not proper length as specified or marked.
Weight	Gross or net weight in excess of specified, or in excess of box certificate weight maximum.

4.3.2 *Inspection of the end item.* Inspection of the end item shall be in accordance with the classification of defects and acceptable quality levels (AQL's) set forth below. Each roll or package of sheets shall constitute one unit of the lot for purposes of determining lot size for sampling.

4.3.2.1 *Examination of the end item for defects in workmanship.* The sample unit for this examination shall be one sheet full size or one yard full width for rolls. Examinaton shall be performed in accordance with table IV.

TABLE IV. *Examination for workmanship*

Examine	Defect
Type	Other than as specified.
Color	Other than as specified.
Finish	Other than high polish, if applicable.
Workmanship	Any dirt, oil, or foreign matter affecting appearance. Any deep gouge or scratch. Any holes (other than allowable pinholes). Any rough or sharp edges. Any cuts, blisters, bubbles, tears, cuts, scuffs, cracks, pits, or creases. Any chipping. Any spot, stain, or other discoloration affecting appearance.

4.3.2.2 *Examination of the end item for defects in dimensions.* The sample unit for this examination shall be one sheet full size, or one complete roll. Examination shall be performed in accordance with table V.

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TABLE V. *Examination for dimensions*

Examine	Defect
Size of sheets	Any variation greater than tolerances specified in contract or order.
Width of roll	Any width less than specified.
	Any width exceeding that specified.
Thickness	Any thickness deviation greater than applicable tolerances specified.

4.3.2.3 *Examination of the end item for content of unit package.* The sample unit shall be one roll or one package of sheets, as applicable. Examination shall be performed in accordance with table VI.

TABLE VI. *Examination for content of unit package*

Examine	Defect
Count per package of sheets	Less than specified or indicated amount.
Length of roll	Less than specified.

4.3.2.4 *Examination of the end item for defects in packaging of sheets and unrolling of rolls.* The sample unit for this examination shall be one package of sheets or one roll. Examination shall be performed in accordance with table VII.

TABLE VII. *Examination for packaging*

Examine	Defect
Unrolling of rolls	Not suitably restrained from unwinding.
	Material blocks to the extent that unrolling causes damage or tearing rendering material unserviceable.
Separation of sheets	Sheets block to the extent that separation causes damage or tearing rendering material unserviceable.

4.3.2.5 *Inspection levels and acceptable quality levels (AQL's) for examinations.* The inspection levels for determining the sample size and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1	L-4	4.0
4.3.2.1	I	2.5
4.3.2.2	L-3	6.5
4.3.2.3	L-4	2.5
4.3.2.4	L-2	6.5

4.3.3 *Certification.* When specified in the contract or purchase order, the contractor shall forward to the procuring agency for each order of sheets a certified statement (1) identifying the plastic material used by manufacturer's name, trade name, and identification number, (2) certifying that the plastic material used is the same as to manufacturing process and composition as that which was found to conform to all requirements of this specification, (3) presenting the results of the material acceptance tests, and (4) certifying that the sheet or film conforms to the requirements of the specification. The report covering the tests shall be referenced on the certified statement.

4.4 **Test procedures.**

4.4.1 *Conditioning test specimens and test conditions.* For those tests where conditioning is required, the specimens shall be conditioned in accordance with Fed. Std. No. 1 and tested under these conditions.

4.4.2 *Thickness.* Thickness measurements shall be made with a micrometer or a suitable calibrated measuring instrument capable of measuring to 0.0001 inch.

4.4.3 *Width and length.* The width and length of the sheet shall be measured with a steel tape or other suitable measuring device.

4.4.4 *Orientation release stress.* The orientation release stress shall be determined by testing five specimens in accordance with ASTM Method D 1504-61.

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4.4.5 *Orientation ratio.* Cut from the sheet or film 3-inch diameter circles and, by any convenient means other than notching the edge, mark the machine direction diameter. A minimum of three specimens, one from the center and one from each edge of the sheet or film, shall be tested. Immerse the specimens in ethylene glycol or other suitable medium proven to have no effect on the material under test for 5 minutes at 130° ± 2°C. (234° ± 4°F.). Measure the dimensional change in both machine and transverse directions to the nearest 0.1 inch. The orientation ratio is the ratio of the original dimension to the final dimension (see note 1).

Note 1. It is not necessary to use a circular specimen for this test. Square specimens with the required circle inscribed on the surface have been successfully used.

4.4.6 *Density.* The density shall be determined by testing three specimens in accordance with ASTM Method D 1505-60T.

4.4.7 *Specific gravity.* Three specimens shall be tested in accordance with Method 5011 or 5012 of Fed. Test Method Std. No. 406.

4.4.8 *Water absorption.* Determine water absorption by testing three specimens in accordance with Procedure A, Method 7031 of Fed. Test Method Std. No. 406.

4.4.9 *Tensile strength and elongation.* These determinations shall be made by testing ten specimens in accordance with Procedure A of Method 1013 in Fed. Test Method Std. No. 406. Cross head shall move at the rate of 1/2 inch per minute.

4.4.10 *Dissipation factor and dielectric constant.* The dissipation factor and dielectric constant shall be determined by testing five specimens, 1/8-inch in thickness, in accordance with Method 4021 of Fed. Test Method Std. No. 406 using the two-electrode system.

4.4.11 *Luminous transmittance.* Determine luminous transmittance by testing three specimens, 1/8-inch in thickness, in accordance with Method 3022 of Fed. Test Method Std. No. 406.

4.4.12 *Water vapor permeability.* Water vapor permeability shall be determined in accordance with Method 7032 of Fed. Test Method Std. No. 406.

4.4.13 *Chemical resistance.* This determination shall be made by testing three specimens in accordance with Method 7011 of Fed. Test Method Std. No. 406.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Rolls.* Film material supplied in rolls shall be wound on a substantial core with an outside diameter of not less than 3 inches. The rolls shall not exceed 150 pounds in weight or 14 inches in diameter, and shall be suitably restrained from unwinding. Each roll shall be wrapped with at least one layer of kraft wrapping paper, conforming to UU-P-268, and tightly sealed with tape conforming to PPP-T-60.

5.1.1.2 *Sheets.* Sheet material shall be interleaved to prevent adherence of sheets to each other and shall be overwrapped with kraft wrapping paper as specified in 5.1.1.1.

5.1.2 *Level C.* Film and sheet material shall be packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 *Level A.*

5.2.1.1 *Rolls.* Each packaged roll of material shall be packed in fiber drums or boxes conforming to type II or III, grade A of PPP-D-723 or class 2, V3c or V3s of PPP-B-636. Closures shall be in accordance with the applicable specification.

5.2.1.2 *Sheets.* Packages of sheet material shall be packed in overseas class or type shipping containers conforming to PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636 (class 2, V3c or V3s) or PPP-B-576. Unless otherwise specified, containers shall be provided with a case liner made of material conforming to UU-P-271 and fabricated in accordance with MIL-L-10547. Case liners for containers conforming to PPP-B-636 shall be omitted provided all seams including manufacturer's joints are sealed with tape conforming to PPP-T-60. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight of wood or wood-cleated containers shall not exceed approximately 200 pounds; fiberboard containers shall not exceed the weight limitations of the applicable container specification.

5.2.2 *Level B.*

5.2.2.1 *Rolls.* Each packaged roll of material shall be packed in fiber drums or boxes conforming to type I, grade A of PPP-D-723 or class 1 of PPP-B-636. Closures shall be in accordance with the appropriate specification.

5.2.2.2 *Sheets.* Packages of sheet material shall be packed in domestic class or type shipping containers conforming to PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-576. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight of wood or wood-cleated containers shall not exceed 200 pounds; fiberboard containers shall not exceed the weight limitations of the applicable container specification.

5.2.3 *Level C.* The material shall be packed to afford adequate protection against damage during shipment within the continental United States direct from the supply source to the first receiving activity. This level, as a minimum, will conform to applicable carrier rules and regulations.

5.3 **Marking.**

5.3.1 *Civil agencies.* Unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following information:

Federal Spec. L-P-506a
Color (if other than colorless)
Finish (if other than high polish)
Size and thickness

5.3.3 *Precautionary marking.*

5.3.3.1 *Rolls of film.* In addition to the marking above, the following precautionary marking shall appear in letters not less than 3/4 inch in height on at least one side, and wherever practical on two sides, of each drum or box containing rolls:

KEEP COOL AND DRY

5.3.3.2 *Flat sheets.* Containers for flat sheets shall be marked on at least one side and wherever practical on two opposite sides, with the following:

KEEP COOL AND DRY

6. NOTES

6.1 **Intended use.** Type I material is intended for general purpose use such as windows in envelope and in carton, in wrapping

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of articles, laminating and the like. Type II material has better elongation properties than type I.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type required (1.2).
(*c*) Form required (3.2).
(*d*) Dimensions required (3.3).
(*e*) Color and transparency required, if any (3.5).
(*f*) Finish desired (3.6).
(*g*) Chemical resistance required, if any (3.7).
(*h*) Whether fog resistance is required (3.9).
(*i*) Certification, if required (4.3.3).
(*j*) Levels of packaging and packing (section 5).

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Plastic sheet or film, not woven, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Plastic film or sheeting, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—EL O E
Navy—YMC

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Copies of this specification may be purchased for 5 cents each.

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